

Mean-Field Dynamics with Channel Mixing via a Fixed Kernel L

Abstract

We study a two-layer mean-field (MF) model with r channel indices and a fixed mixing kernel L that couples channels into the second-layer pre-activations. We formalize the MF ODEs for the first- and second-layer weights, introduce ψ_2 -type sub-Gaussian control to obtain a priori bounds, and establish well-posedness (existence and uniqueness) via a Picard iteration. We also discuss the interpretation of the dynamics and the role of the kernel L .

1 Setup and mean-field equations

Let $Z = (X, Y) \sim \mathcal{P}$ denote the data distribution on $\mathcal{X} \times \mathcal{Y}$. The input $X \in \mathbb{R}^d$ and the label $Y \in \mathbb{R}$. Let Ω_1, Ω_2 be Polish spaces with probability measures ρ^1, ρ^2 , and write $C_1 \sim \rho^1$, $C_2 \sim \rho^2$. Fix an integer $r \geq 1$ (the number of channels). We consider: - a feature map $f_1 : \Omega_1 \rightarrow \mathbb{R}^d$, - activations $\varphi_1, \varphi_2 : \mathbb{R} \rightarrow \mathbb{R}$, with φ'_2 the derivative, - learning-rate schedules $\xi_1, \xi_2 : \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}_{\geq 0}$, - a measurable mixing kernel $L : \Omega_2 \times \{1, \dots, r\} \rightarrow \mathbb{R}$, where we denote $L_{c_2, k} = L(c_2, k)$.

The trainable MF parameters are

$$w_1 : \mathbb{R}_{\geq 0} \times \Omega_1 \times \{1, \dots, r\} \rightarrow \mathbb{R}, \quad w_2 : \mathbb{R}_{\geq 0} \times \Omega_2 \rightarrow \mathbb{R}.$$

Define the second-layer pre-activation for $c_2 \in \Omega_2$ by

$$H_2(t, c_2; X, W) = \sum_{k=1}^r L_{c_2, k} m_k(t; X, W), \quad m_k(t; X, W) \equiv \mathbb{E}_{C_1} [w_1(t, C_1, k) \varphi_1(\langle f_1(C_1), X \rangle)]. \quad (1)$$

The model output is

$$\hat{y}(X; W(t)) = \mathbb{E}_{C_2} [w_2(t, C_2) \varphi_2(H_2(t, C_2; X, W(t)))]. \quad (2)$$

Let $\mathcal{L} : \mathcal{Y} \times \mathbb{R} \rightarrow \mathbb{R}_{\geq 0}$ be the loss, and write the output derivative as

$$d_L(Z; W) \equiv \partial_2 \mathcal{L}(Y, \hat{y}(X; W)). \quad (3)$$

The MF dynamics are given by the following ODEs (for $c_1 \in \Omega_1$, $c_2 \in \Omega_2$, $k \in \{1, \dots, r\}$):

$$\partial_t w_1(t, c_1, k) = -\xi_1(t) \mathbb{E}_Z [d_L(Z; W(t)) \varphi_1(\langle f_1(c_1), X \rangle) \mathbb{E}_{C_2} [L_{C_2, k} \varphi'_2(H_2(t, C_2; X, W(t))) w_2(t, C_2)]] , \quad (4)$$

$$\partial_t w_2(t, c_2) = -\xi_2(t) \mathbb{E}_Z [d_L(Z; W(t)) \varphi_2(H_2(t, c_2; X, W(t)))]. \quad (5)$$

Initial conditions are $w_1(0, \cdot, \cdot) = w_1^0(\cdot, \cdot)$ and $w_2(0, \cdot) = w_2^0(\cdot)$.

Remark 1 (Backprop structure). Equation (5) is the usual second-layer MF update. Equation (4) shows that each channel k receives a backpropagated signal

$$b_k(t) = \mathbb{E}_{C_2} [L_{C_2, k} \varphi'_2(H_2(t, C_2; X, W(t))) w_2(t, C_2)] ,$$

so that $\partial_t w_1(t, c_1, k)$ is proportional to $\mathbb{E}_Z [d_L(\cdot) \varphi_1(\langle f_1(c_1), X \rangle) b_k(t)]$. The kernel L mixes the r channel summaries m_k into each $H_2(\cdot)$.

2 Assumptions

Assumption 1 (Learning-rate, regularity, inputs, mixing). *There exists $K \geq 1$ such that:*

- ξ_1, ξ_2 are bounded on compact intervals and piecewise continuous.
- $\varphi_1, \varphi_2, \varphi'_2$ are K -bounded and K -Lipschitz; moreover φ'_2 is non-vanishing.
- $\partial_2 \mathcal{L}(y, \cdot)$ is K -bounded and K -Lipschitz for all y in the support of $\mathcal{P}$.
- X is K -sub-Gaussian and $\|f_1(c_1)\| \leq K$ for all c_1 ; hence $\langle f_1(c_1), X \rangle$ is K -sub-Gaussian.
- The mixing kernel satisfies $\sup_{c_2, k} |L_{c_2, k}| \leq K$ and $k \mapsto L_{c_2, k}$ is measurable for each c_2 .

Assumption 2 (Sub-Gaussian initialization). *There exists $K \geq 1$ such that*

$$\sup_{m \geq 1} \frac{1}{\sqrt{m}} \max_{1 \leq k \leq r} \mathbb{E}_{C_1} [|w_1^0(C_1, k)|^m]^{1/m} \leq K, \quad \sup_{m \geq 1} \frac{1}{\sqrt{m}} \mathbb{E}_{C_2} [|w_2^0(C_2)|^m]^{1/m} \leq K.$$

3 Norms and ψ_2 control

For $t \geq 0$, define the L^{50} -type sup-in-time norms

$$\|w_1\|_t \equiv \max_{1 \leq k \leq r} \mathbb{E}_{C_1} \left[\sup_{s \leq t} |w_1(s, C_1, k)|^{50} \right]^{1/50},$$

$$\|w_2\|_t \equiv \mathbb{E}_{C_2} \left[\sup_{s \leq t} |w_2(s, C_2)|^{50} \right]^{1/50}, \quad \|W\|_t \equiv \max(\|w_1\|_t, \|w_2\|_t).$$

We also define ψ_2 -type sub-Gaussian seminorms (calibrated at $m = 50$):

$$w_{1\psi, t} \equiv \sqrt{50} \sup_{m \geq 1} \frac{1}{\sqrt{m}} \max_{1 \leq k \leq r} \mathbb{E}_{C_1} \left[\sup_{s \leq t} |w_1(s, C_1, k)|^m \right]^{1/m},$$

$$w_{2\psi, t} \equiv \sqrt{50} \sup_{m \geq 1} \frac{1}{\sqrt{m}} \mathbb{E}_{C_2} \left[\sup_{s \leq t} |w_2(s, C_2)|^m \right]^{1/m}, \quad W_{\psi, t} \equiv \max(w_{1\psi, t}, w_{2\psi, t}).$$

By choosing $m = 50$, we have $W_{\psi, t} \geq \|W\|_t$.

4 Well-posedness (existence and uniqueness)

Define the solution operator F for any $W' = (w'_1, w'_2)$:

$$F_1(W')(t, c_1, k) = w_1^0(c_1, k) - \int_0^t \xi_1(s) \mathbb{E}_Z [d_L(Z; W'(s)) \varphi_1(\langle f_1(c_1), X \rangle) B_k(s; X, W'(s))] ds,$$

$$F_2(W')(t, c_2) = w_2^0(c_2) - \int_0^t \xi_2(s) \mathbb{E}_Z [d_L(Z; W'(s)) \varphi_2(H_2(s, c_2; X, W'(s)))] ds,$$

where

$$B_k(s; X, W') \equiv \mathbb{E}_{C_2} \left[L_{C_2, k} \varphi'_2(H_2(s, C_2; X, W')) w'_2(s, C_2) \right].$$

We look for fixed points $W = F(W)$.

Theorem 1 (Global well-posedness). *Under Assumptions 1–2, for any initialization $W(0) = (w_1^0, w_2^0)$ with $W_{\psi, 0} < \infty$, the MF ODEs (4)–(5) admit a unique global solution $W(t)$ on $[0, \infty)$ with $W_{\psi, t} < \infty$ for each t .*

Proof sketch. One proves a priori bounds of the form $\|W\|_t \leq K_0(t)$ and $W_{\psi,t} \leq K_0(t)$ with $K_0(t)$ polynomial in t and $W_{\psi,0}$, using the bounded/Lipschitz properties and sub-Gaussianity. Then one shows that F maps the ball $\{W' : W'_{\psi,T} \leq K_0(T)\}$ into itself, and establishes a difference inequality

$$\|F(W') - F(W'')\|_t \leq C \int_0^t \left((1+B) \|W' - W''\|_s + e^{-cB^2} \right) ds$$

for arbitrary $B \geq 0$ (good/bad event decomposition from sub-Gaussian tails). Iterating and choosing $B = \sqrt{m}$ yields convergence of the Picard iterates $F^{(m)}(W')$ in $\|\cdot\|_T$, and uniqueness follows by the same estimate. Details follow the standard MF ODE strategy with ψ_2 control.

5 Technical roadmap: key lemmas and intuition (2-layer case)

This section condenses the existence/uniqueness strategy (as in the multilayer analysis) into the essential steps needed for the two-layer system (4)–(5). Each lemma below states what to prove and why it is true; the proofs mirror the multilayer case but simplify because only the top layer backpropagates.

Notation map (from multilayer to this paper). - $W = (w_1, w_2)$ here corresponds to the parameter collection across layers.

- H_2 here plays the role of the top-layer pre-activation; H_i for $i < 2$ is not needed explicitly.
- Differences of drifts noted as $\Delta^{(2)}$ and $\Delta^{(1)}$ here correspond to the gradient integrands for w_2 and w_1 in the ODEs (5) and (4), respectively.
- The ψ_2 control $\cdot_{\psi,t}$ is defined to dominate the L^{50} sup-in-time norms $\|\cdot\|_t$.

Lemma A (A priori ψ_2 and L^{50} bounds). There is a nondecreasing function $K_0(t)$, polynomial in t and $W_{\psi,0}$, such that for all $t \geq 0$,

$$\|W\|_t \leq K_0(t), \quad W_{\psi,t} \leq K_0(t),$$

and, moreover, for any $B \geq 0$,

$$\mathbb{P} \left(\max \left\{ \sup_{s \leq t} \sup_{c_2} |w_2(s, c_2)|, \sup_{s \leq t} \sup_{c_1, k} |w_1(s, c_1, k)| \right\} \geq K_0(t) B \right) \leq C e^{-cB^2}.$$

Intuition. For layer 2, the drift is $-\xi_2 \mathbb{E}[d_L \varphi_2(H_2)]$; boundedness/Lipschitz of $\partial_2 \mathcal{L}$ and φ_2 yields linear-in-time growth in moments. For layer 1, the drift is $-\xi_1 \mathbb{E}[d_L \varphi_1(\langle f_1, X \rangle) B_k]$, where $B_k = \mathbb{E}_{C_2}[L_{C_2,k} \varphi'_2(H_2) w_2]$ is bounded in moments by the layer-2 controls and ψ_2 stability under Lipschitz maps. Using Hölder/Cauchy–Schwarz and Grönwall-type arguments gives polynomial $K_0(t)$. Sub-Gaussian tails follow from the ψ_2 bounds and a union bound on the sup in time.

Lemma B (Invariance: F maps the ψ_2 -ball into itself). For any $T > 0$, define the set

$$\mathcal{W}_T^0 \equiv \left\{ W' : W'_{\psi,T} \leq K_0(T), \mathbb{P} \left(\max_{s \leq T} \max \{ |w'_1(s)|, |w'_2(s)| \} \geq K_0(T) B \right) \leq C e^{-cB^2}, W'(0) = W(0) \right\}.$$

Then $F(\mathcal{W}_T^0) \subseteq \mathcal{W}_T^0$. *Intuition.* The integral form defining F applies bounded/Lipschitz drifts to W' , so the same moment and tail bounds propagate with constants controlled by $K_0(T)$.

Lemma C (Forward Lipschitz). For $t \leq T$ and any $W', W'' \in \mathcal{W}_T$, we have

$$\mathbb{E} \left[\sup_{s \leq t} \sup_{c_2} |H_2(s, c_2; X, W') - H_2(s, c_2; X, W'')|^2 \right]^{1/2} \leq C K_0(T) \|W' - W''\|_t,$$

and

$$\sup_{s \leq t} \text{ess-sup}_X |\hat{y}(X; W'(s)) - \hat{y}(X; W''(s))| \leq C K_0(T) \|W' - W''\|_t.$$

Intuition. H_2 is linear in w_1 via m_k and L , and $\hat{y}$ is linear in w_2 and Lipschitz in H_2 ; bounded L , φ_1 , φ_2 transfer differences in (w_1, w_2) linearly.

Lemma D (Backward Lipschitz under a good event). Fix $B \geq 0$ and suppose both W', W'' satisfy

$$\mathbb{P} \left(\max_{s \leq T} \max\{|w'_1|, |w'_2|\} \geq K_0(T)B \right), \mathbb{P} \left(\max_{s \leq T} \max\{|w''_1|, |w''_2|\} \geq K_0(T)B \right) \leq C e^{-cB^2}.$$

Then, for $t \leq T$,

$$\mathbb{E} \left[\sup_{s \leq t} \sup_{c_2} |\Delta^{(2)}(s, c_2; W') - \Delta^{(2)}(s, c_2; W'')|^2 \right]^{1/2} \leq C K_0(T)^2 \left((1+B) \|W' - W''\|_t + e^{-cB^2} \right),$$

and similarly for the first-layer drift difference $\Delta^{(1)}$. *Intuition.* On the good event $\{\max |w| \leq K_0(T)B\}$, the derivatives of the loss and activations keep the drift Lipschitz in W , with a $(1+B)$ factor. The bad event contributes an exponentially small remainder thanks to sub-Gaussian tails.

Lemma E (Difference inequality for the solution operator). Combining Lemmas C–D and integrating in time yields, for $t \leq T$,

$$\|F(W') - F(W'')\|_t \leq C K_0(T)^2 \int_0^t \left((1+B) \|W' - W''\|_s + e^{-cB^2} \right) ds.$$

Intuition. The drift differences control the integral differences (by Minkowski), leading to a Grönwall-like inequality with a tunable parameter B that trades Lipschitz strength for a small exponentially decaying error.

Proposition (Picard iteration and uniqueness). Iterate Lemma E: for $m \geq 1$,

$$\|F^{(m)}(W') - F^{(m)}(W'')\|_T \leq \frac{(CT(1+B))^m}{m!} \|W' - W''\|_T + C e^{CT(1+B)} e^{-cB^2}.$$

Choosing $B = \sqrt{m}$ makes the remainder e^{-cm} dominate the combinatorial growth, forcing convergence of $F^{(m)}(W')$ to a fixed point in $\|\cdot\|_T$. Uniqueness follows by letting $m \rightarrow \infty$ in the same bound with $W'' = F(W')$.

Two-layer simplifications (vs. multilayer). - No deep backward induction: the base case is the top layer (w_2), and the first layer only depends on that base case. Constants are $K^{O(1)}$ rather than $K^{O(L)}$. - Forward Lipschitz: H_2 depends linearly on w_1 through bounded L and φ_1 , and $\hat{y}$ depends linearly on w_2 ; this eliminates layer-wise products. - Backward Lipschitz: only the pair $(\Delta^{(2)}, \Delta^{(1)})$ matters. Good-event control uses the same ψ_2 tails; there is no accumulation over depth.

6 Discussion and interpretation

- **Channel mixing via L .** The kernel L mixes the r channel summaries m_k into each $H_2(\cdot)$, creating a low-rank structure at the level of pre-activations. The first-layer dynamics for channel k are driven by the backprop signal B_k formed by $L_{\cdot,k}$, φ'_2 , and w_2 .

- **Consistency of gradients.** The factor $d_L(Z; W)$ is the global error signal (derivative of loss w.r.t. output). The second-layer update (5) is standard; the first-layer update (4) inserts the Jacobian chain $L \varphi'_2(\cdot) w_2$, averaged over C_2 .

- **Identifiability and scaling.** If columns $k \mapsto L_{\cdot,k}$ are nearly collinear, the m_k may not be individually identifiable; the dynamics can still progress but may converge to equivalent representations. Normalizing L (e.g., $\sup_{c_2} \sum_k L_{c_2,k}^2 \leq 1$) helps constants.

- **Why ψ_2 ?** Sub-Gaussian control is stable under Lipschitz maps and expectations, enabling uniform-in-time bounds and a good/bad event decomposition that closes the Picard iteration globally in time.

- **Extensions.** Biases, noise (diffusion), and finite-width couplings can be added with analogous definitions. Convergence to stationary points requires additional structural assumptions (e.g., strict convexity in an appropriate representation).